



Seagate Technologies has announced shipping of the Cheetah® 15K.7 enterprise-class hard drives into the worldwide distribution channel

LCD Monitor Test

nothing less than a 22-inch monitor.

This year we considerably ramped up our test process. Firstly, we used a display calibrator, the DataColor Spyder 3 Elite and each of the monitors was carefully calibrated before testing them. We decided not to test 14, 15, and 17-inch LCDs this year since 19-inch displays have become so affordable. A trend we hope to see continue was the presence of larger panels, namely the 24- and 26-inch categories. Last year we only received one 24-inch display; this year we came across three of them. TN panels abound because of their cheaper price point but there are still a few high-end S-IPS and PVA panels floating around and although we weren't able to get our hands on all of them, we did have some participation from this category.

26 and 24-inch LCDs – Glory be

These two categories represent the top of the pyramid of LCD monitors available today. This size category is also where one would find the high-end PVA and IPS LCD panels, and in general, the best specifications. It's also the costliest of the categories in test, and in general a 24-inch display will cost you at least Rs. 15,000, while a high-end monitor could cost upward of Rs. 75,000. One of the most pleasing things about these two size categories is the resolution – a whopping 1920 x 1200 pixels; this is even higher than the 1080p HD resolution that equates to 1920 x 1080 pixels. The bigger viewing area is another plus. Obviously, such a resolution is just what the doctor prescribed for gamers who want all the details at higher resolutions but one also needs a super graphics card to pull off gaming at this resolution in high detail. Image editing professionals also look at higher-end panels in these size categories. Do note since both 24- and 26-inch displays pack the same number of pixels



AOC 2434 PW

on-screen, the pixel pitch on a 26-inch display is larger and for some the 24-inch may look marginally crisper due to the pixels being grouped closer together, although our eyes could not discern significant difference.

Features

NEC's MultiSync 2690WUXi sports a smooth satin-like finish and is a relief from the other piano-glossy finished panels.

HOW WE TESTED

We divided the monitors received into categories on the basis of their screen size. On the features front, we logged all the physical specifications of the LCDs including details on the panel, the weight and dimensions and connectivity. We also rated the build quality, menu system and bezel thickness of each panel.

Our test system consisted of an Intel Core 2 Duo E6700 CPU (2.66 GHz), 4 GB DDR2 800 MHz RAM and a ZOTAC GeForce GTX 295 graphics card. We used the Spyder 3 Elite to calibrate each display and also give us the real contrast ratio and luminance figures. For luminance we kept the brightness at 100 per cent for all displays. For calibration and testing the contrast ratio, both the contrast and brightness level were set at 50 per cent. Post calibration we ran Display Mate to judge the contrast ratio for ourselves via the grey scale intensity test; this consists of checking visibility of grey squares of varying intensity on a black background.

Display Mate's colour test suite allowed us to check for colour gamut and shade

intensity rendition for each display. In the colour spectrum test we look for the wideness of the colour gamut where the primary colours are separated by the secondary colours and the entire spectrum is presented on-screen. Colours should smoothly merge into others without vertical lines called "banding". When testing for purity of the

primary colours and individual shades we use the 16-shade ramp test and the 256-ramp test. 16 and 256 bands (respectively) are presented for each of the primary and secondary colours one colour at a time and these bands vary equally in intensity moving from highest intensities to lowest intensities. We check for uniformity of change



in intensity and adequate separation between each band – this tells us that the monitor is able to display variable intensities of a particular shade well. Generally, IPS and PVA panels that are natively 8-bit are better performing than 6-bit TN panels that use dithering to make up for additional colour intensities that they cannot otherwise produce.

swivel and tilt mechanisms work very well – great quality of movable parts. The NEC AccuSync 24WMCX has a decidedly value look to it, and it's evident this one is built for relatively conservative users – we say relatively because this cheaper display still costs



NEC MultiSync LCD 2690 WUXi

Our movie test consisted of a set of four HD 1080p clips encoded in WMV format. A mix of natural scenery, fast action scenes and underwater photography, these video clips allow an objective look at the colour rendition, contrast and brightness level of a display. We used F.E.A.R and Crysis for our game tests. Both these games are studies in opposites — Crysis with its realistic scenery, bright outdoors and lighting depicting an island paradise while F.E.A.R with its cramped, dimly lit corridors and slow motion action scenes. Using these two games, detail, contrast, brightness, colour rendition and viewing angles are a snap to assess.